



● Jianhao Jia

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Educational Background

ITCS, Shanghai University of Finance and Economics

2021.9--now

- Advisor: Prof. Nick Gravin.
- Research focus: Algorithmic Game Theory, Online Algorithms, Mechanism Design without Money.
- Thesis topic: Strategic Decision Making.

Pilot Class, RIIS, Shanghai University of Finance and Economics

2017.9--2021.6

- RIIS(Research Institute for Interdisciplinary Science). Majoring in interdisciplinary disciplines, involving computer science, finance, statistics etc.

Research Interests

Mechanism Design without Money

- Strategic facility location, strategic clustering, and combinatorial social choice.

Online Algorithms

- Prophet inequalities, online matching, and resource allocation under Markovian demands.

Learning-Augmented Mechanisms

- Mechanism design with predictions, consistency and robustness trade-offs.

Publications

Approximation guarantees of median mechanism in R^d J. Jia, N. Gravin

STOC 2025, Prague, Czech

- Resolved a longstanding open problem regarding the tight approximation ratio of the median mechanism in multi-dimensional L_q normed spaces.
- Extended results to the mechanism design with predictions framework.

Online resource allocations in Markov Chains J. Jia, H. Li, K. Liu, Z. Liu, J. Zhou, N. Gravin, Z.G. Tang

WWW 2023, Austin, USA

- Established a new competitive ratio framework for allocation problems where future demand is correlated with the current state.
- Designed algorithms recovering the standard 1/2 prophet inequality ratio for single-item settings and $1 - o(1)$ ratio for multi-item settings under Markovian demand evolution and mild assumptions.

Under Submission

- Facility Location with k-means+ +: Incentives and Resource Augmentation

Working papers

- **k-facility location: How to close facility strategically?**

Industry Experience (Internship)

Ant Finance, Alibaba Co.

2022.4 - 2023.2

- Collaborated on the project "Online resource allocations in Markov Chains" to model the evolution of financial markets and user behavior.
- Conducted theoretical analysis and applied the Markovian allocation model to large-scale company datasets, demonstrating improved efficiency over standard baselines.
- Resulted in a full paper publication at **WWW 2023**.

Guotai Junan Securities Co.

2019.7 - 2019.8 & 2020.7-2020.8

- Developed Python crawlers to automate data collection from web sources and integrated financial data.
- Performed quantitative analysis on fund products and manager performance using Python.

Service

- TA: Discrete Math (2022, 2024, 2025); Online Decision Making (2024).
- Reviewer: MOR, TCS, AAMAS, SODA; PC member: EC 2026.
- Programming skills: python, MATLAB, etc.; Language: English (Proficient- TOEFL 104), Mandarin (Native).